

Fitz Roy Geology

Tectonic History, Field Observations, and Hypothesis Development in Patagonia

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Overview

The study site is located in the northern region of Parque Nacional Los Glaciares, Patagonia, Argentina (approximately 49.28°S, 72.94°W), on the northern Patagonian Andes, east across a valley from the Fitz Roy massif and lagoons. It was observed from the Laguna de los Tres hiking trail.

From a tectonic perspective, the region has undergone a complex evolution. The subduction of the Nazca Plate beneath the South American Plate introduced crustal shortening, uplift, and magmatism, forming the Patagonian Andes over the Cenozoic (De Arellano, 2009) and producing the batholithic and metamorphic rocks visible today over tens of millions of years (Hervé et al., 2007).

Beneath the northern Patagonian segment of the Andes lies the North Patagonian Batholith, a large body of ancient igneous rock composed mostly of granite and similar rocks that formed underground as magma cooled between roughly 173 and 5 million years ago (Hervé et al., 2007). These rocks were part of the Andean volcanic arc, created by the subduction of the Nazca Plate.

Over time, these bodies and neighboring metamorphic rocks became the foundation of the mountains of Parque Nacional Los Glaciares, including Fitz Roy. The crust was later uplifted and deformed as the Andes continued to grow. During the Miocene to the present, uplift may have accelerated due to changes in the angle of the subducting slab and possibly from isostatic rebound after glaciers melted and reduced surface weight (De Arellano, 2009). These processes produced the steep, high-relief landscapes of northern Patagonia.

The region's tectonic history is also shaped by back-arc extension, slab rollback, and the collision of the Chile Ridge with the trench, all of which affected magma generation and mountain building (Ramos & Ghiglione, 2008). Many peaks and valleys result from uplift driven by tectonic forces combined with glacial erosion over millions of years.

Observations

Figures 1 and 2 show photos taken from the trail on the east side of the valley, looking across to the rocky massif. Figure 3 is a hand-drawn sketch of the study site. Figure 4 shows a Google Earth satellite view of the study area; the red curve marks the top of the study site, located east of the main valley. To the west lie several glacier-fed lagoons, including Laguna Sucia (the large blue lake at lower left in Figure 4). The light-gray belt extending through the center of the valley represents the active meltwater channel. Vegetation is more condensed on the flatter valley floor, while the upper slopes are bare rock.



Figure 1: Most zoomed-in photo of the study site.



Figure 2: Medium zoomed-in photo of the study site.

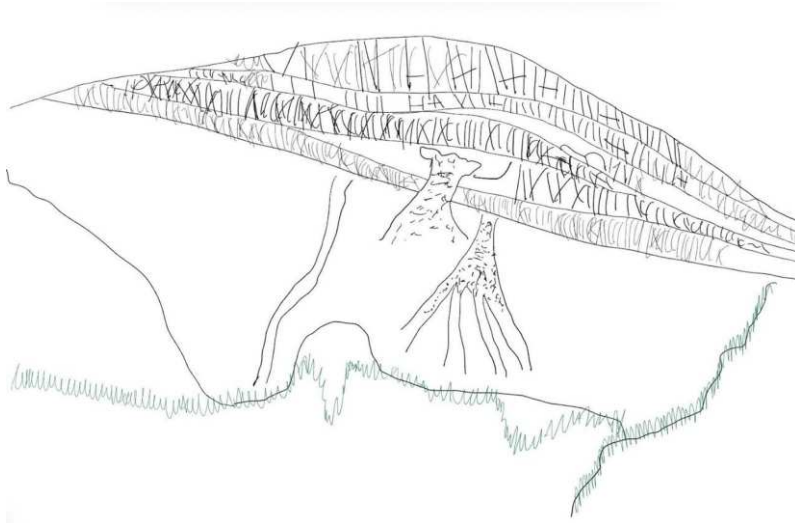


Figure 3: Hand-drawn sketch of the study site.



Figure 4: Google Earth satellite view of the study area (Image © 2025 Airbus). The red curve marks the top of the study site. Laguna Sucia is visible at lower left; the light-gray belt through the center of the valley is the active meltwater channel.

The massif runs roughly southwest-northeast, parallel to the valley (see Figure 4), where the floor is forested and the higher elevations are mostly bare rock.

The cliff face displays multiple light- to tan-colored rock layers that appear bedded and sub-parallel over hundreds of meters, with a gentle tilt dipping slightly toward the right side of the photo frame (see Figures 1 and 2). Layer thickness varies across the massif,

suggesting differential erosion between more and less resistant rock. Vertical to sub-vertical fractures cut across the rock layers, with talus collections visible at the base of the cliff (see Figure 3).

The Google Earth view (Figure 4) shows the light-gray channel belt at the valley bottom, replenished by streamwater from the Laguna Sucia outlet area, likely meltwater from the Southern Patagonian Ice Field. Downstream, multiple abandoned or secondary channels are visible alongside fan-shaped sediment deposits.

Hypothesis Development

Two hypotheses are developed, each with testable predictions.

Hypothesis 1: If the rocks in the study massif were compressed during the evolution of the Andean mountains, then the parallel alignment of rock layers was later tilted, and fractures opened as the massif kept uplifting, because subduction shortens and cracks the crust.

Predictions:

- (a) Most surfaces on the cliff will share a similar dip to that of the regional Andean trend.
- (b) Talus at the base of the cliff will consist of young pieces of the same rock types as the uppermost sections of the cliff, showing recent fractures in those areas.
- (c) Fractures will be vertical or near-vertical and cut across the parallel rock layers, with the same fracture orientations repeated along the cliff face.

Hypothesis 2: If meltwater from the nearby glacier and lagoons carries a large volume of silt and gravel, then it spreads out on the valley floor and builds the wide, light-gray belt seen in Figure 4.

Predictions:

- (a) The stream appears milky on warm afternoons and clearer on cold mornings, because glacier meltwater contains suspended silt.
- (b) Young gravel bars appear after hot days or storms at the valley bottom.
- (c) Sediment is coarser and more angular near the glacier and becomes finer and rounder downstream.
- (d) The gray belt widens where the stream's slope decreases and narrows where stream velocity increases.

Discussion

Vegetation patterns observed at the site reflect the same processes outlined in the hypotheses. Forest covers the valley floor, where the surface is flatter and soil is better supported. The steep, fractured cliff above lacks trees because the soil is thin and slides off with rockfall.

At the valley bottom, seasonal meltwater floods bring new sediment. Plants colonize older, higher surfaces that are disturbed less frequently, while seedlings on lower surfaces

are uprooted when the stream shifts. This creates a positive feedback loop: more plants stabilize surfaces and trap fine sediment, which supports additional plant growth.

These causes also operate across multiple scales. At the mineral level, granite breaks down into sand and silt that retain water. At the landform level, talus and streams control where soil can accumulate. At the climate level, seasonal melt determines the timing and intensity of disturbance. These levels interact: warming promotes ice melt, increasing stream velocity, which reshapes gravel surfaces and can uproot young plants, resetting the cycle.

References

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